Hybrid Video Provenance Framework (HVPF)

Phase 2 Results

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Abstract

This report presents the Phase 2 results of the HVPF pipeline: a three-class decision (organic / manipulated / synthetic) for video clips, compared with the conventional PRNU method (PCE rule). Phase 2 extends Phase 1 with the full Video-ACID dataset, FloreView as an unseen-camera test set, additional genuine re-encodes, and WhatsApp-/YouTube-like copies of genuine, attacked and AI-generated clips. All numbers are generated directly from the result files.

1 Key results

- **Best model:** XGBoost reaches PVA 0.785, macro F1 0.792 and authentic-vs-not AUC 0.899 (EER 0.193) in 5-fold cross-validation on cameras never seen in training, against AUC 0.635 (EER 0.401) for the conventional PCE rule (RQ3).
- **Honest benchmark:** Phase 1 recognised 99% of genuine WhatsApp copies, but only because no fake clip was WhatsApp-compressed. With WhatsApp-like copies of fakes the rate is 57%, which is why Phase 2 scores are lower.
- **Genuine re-encodes:** in the encoder-controlled setup, XGBoost macro F1 rose from 0.703 (Phase 1) to 0.770.
- **5 vs 10 frames:** 10 frames looks better on the full data, but only because sequence models learn the clip length (attacked clips have 30 frames, ACID and AI clips about 6). With clips of equal length the two windows perform the same (Table 5); 5 frames is the main setting.
- **Unseen cameras:** on FloreView (45 cameras, never trained on) genuine videos of phones with a usable PRNU are mostly recognised, but phones whose stabilisation removes the PRNU are the hardest case (Table 7).
- **Features (RQ1):** removing Noiseprint statistics costs the most, followed by PRNU and TSNCS; FFT and residual moments add almost nothing (Figure 2).

2 Data and protocol

The evaluation uses 24,599 clips. Model inputs come from the first five I-frames of each clip. Results are averaged over five grouped cross-validation folds: no camera model is in both training and test. *Realistic* uses all clips; *encoder-controlled* uses only clips encoded by our libx264 pipeline (genuine re-encodes, attacks, AI clips), so encoder traces cannot separate the classes. *Cross-dataset* is FloreView (genuine) plus GenBuster (unseen AI generators), tested with the fixed-split models.

Table 1: Clips per class and source.

Class	Source	Clips
manipulated	VISION-attack	7,200
organic	ACID	4,380
organic	ACID-attack	3,168
organic	FloreView	1,716
organic	VISION	1,259
organic	VISION-attack	2,400
synthetic	GenBuster	1,800
synthetic	GenVidBench	2,676

3 Main results

Table 2: Realistic setup, 5-fold cross-validation (mean ± std over folds), 5 frames. PVA = share of clips assigned to their true provenance class; AUC/EER for genuine vs not genuine.

Model	PVA	Macro F1	AUC (3-class)	AUC genuine vs not	EER
PCE rule	–	–	–	0.635	0.401
SVM	0.761 ± 0.022	0.770 ± 0.028	0.897	0.867	0.215
XGBoost	0.785 ± 0.027	0.792 ± 0.028	0.923	0.899	0.193
GRU	0.774 ± 0.020	0.785 ± 0.021	0.917	0.891	0.205
1D-CNN	0.765 ± 0.036	0.774 ± 0.042	0.909	0.880	0.211
Transformer	0.761 ± 0.018	0.775 ± 0.016	0.908	0.880	0.213

Table 3: Encoder-controlled setup (5-fold) and cross-dataset test (FloreView + GenBuster), 5 frames.

	Encoder-controlled			Cross-dataset		
Model	PVA (%)	Macro F1	AUC	PVA (%)	Macro F1	AUC
PCE rule	–	–	0.598	–	–	0.654
SVM	74	0.757	0.841	48	0.583	0.729
XGBoost	75	0.770	0.867	58	0.682	0.886
GRU	74	0.765	0.857	60	0.692	0.842
1D-CNN	74	0.757	0.854	56	0.672	0.848
Transformer	73	0.757	0.848	52	0.653	0.829

Table 4: Phase 1 vs Phase 2 (realistic, 5-fold, 5 frames). Phase 2 is harder by design: fakes also appear as platform-like copies and there are four times more genuine ACID clips.

	PVA (%)		Macro F1		AUC genuine vs not	
Model	Phase 1	Phase 2	Phase 1	Phase 2	Phase 1	Phase 2
PCE rule	–	–	–	–	0.570	0.635
XGBoost	83	78	0.842	0.792	0.937	0.899
GRU	84	77	0.852	0.785	0.930	0.891
Transformer	82	76	0.830	0.775	0.916	0.880

4 5 vs 10 frames

The length-matched columns use only clips with at least 10 I-frames (2,491 test clips per fold), so both windows see the same clips and the clip length carries no information.

Table 5: Effect of the frame window (realistic, 5-fold).

	All clips				Length-matched (≥ 10 frames)			
	5 frames		10 frames		5 frames		10 frames	
Model	PVA	AUC	PVA	AUC	PVA	AUC	PVA	AUC
XGBoost	78	0.899	81	0.915	73	0.831	73	0.845
GRU	77	0.891	82	0.926	71	0.814	71	0.817
1D-CNN	77	0.880	83	0.930	68	0.790	69	0.807
Transformer	76	0.880	83	0.929	70	0.796	70	0.800

5 Results per condition (RQ2)

Share of clips assigned to their true class (%), realistic setup, 5-fold, 5 frames. Values below 70% are in bold. The PCE rule only decides genuine vs not genuine: its high rates on attacks and AI clips come with rejecting most genuine clips.

Table 6: Detection rate per condition.

Condition	n	PCE	SVM	XGB	GRU	CNN	Transf.
Genuine (organic)							
ACID native	4380	36	91	94	93	92	92
VISION native	425	39	78	88	83	83	79
VISION real WhatsApp copy	423	5	**47**	**57**	**46**	**47**	**35**
VISION real YouTube copy	411	10	**57**	**68**	**64**	**62**	**63**
ACID genuine re-encode, CRF 18	528	28	88	92	92	88	89
ACID genuine re-encode, CRF 20	528	25	86	89	88	84	86
ACID genuine re-encode, CRF 20 + WA-like	528	0	**52**	**63**	**38**	**51**	**29**
ACID genuine re-encode, CRF 20 + YT-like	528	8	81	86	84	77	81
ACID genuine re-encode, CRF 23	528	20	84	86	85	81	85
ACID genuine re-encode, CRF 28	528	9	85	86	89	83	85
VISION genuine re-encode, CRF 18	400	24	**67**	74	**69**	70	**67**
VISION genuine re-encode, CRF 20	400	21	**55**	**63**	**59**	**59**	**58**
VISION genuine re-encode, CRF 20 + WA-like	400	2	**25**	**34**	**28**	**29**	**21**
VISION genuine re-encode, CRF 20 + YT-like	400	10	**40**	**48**	**49**	**47**	**45**
VISION genuine re-encode, CRF 23	400	17	**48**	**62**	**52**	**52**	**50**
VISION genuine re-encode, CRF 28	400	9	**43**	**53**	**53**	**50**	**45**
Manipulated							
denoise_a1	400	84	**66**	**57**	**68**	**63**	**66**
denoise_a2	400	84	78	70	77	75	76
denoise_a2 + WA-like	400	99	72	**68**	**67**	**67**	77
denoise_a2 + YT-like	400	95	**68**	**63**	**67**	70	71
denoise_a4	400	88	83	79	82	81	84
injection_a0.25	400	83	70	**67**	**68**	**69**	**66**
injection_a0.5	400	30	78	83	82	77	80
injection_a0.5 + WA-like	400	98	74	70	73	**68**	79
injection_a0.5 + YT-like	400	74	73	**68**	71	71	70
injection_a1	400	2	92	98	96	93	96
removal_a0.25	400	87	**51**	**44**	**49**	**48**	**47**
removal_a0.5	400	96	**66**	75	73	73	73
removal_a0.5 + WA-like	400	100	70	**66**	**68**	**64**	75
removal_a0.5 + YT-like	400	98	**68**	**65**	**64**	**64**	**65**
removal_a1	400	100	86	94	95	95	94
removal_opt	400	99	**62**	**61**	**64**	**66**	**65**
removal_opt + WA-like	400	100	73	**66**	**68**	**66**	74
removal_opt + YT-like	400	100	**62**	**56**	**58**	**60**	**60**
AI-generated (synthetic)							

Condition	n	PCE	SVM	XGB	GRU	CNN	Transf.
CogVideo	294	100	97	98	98	96	97
CogVideo + WA-like	294	100	94	98	98	96	97
CogVideo + YT-like	294	100	95	98	97	95	95
Kling	266	100	89	91	89	85	85
Kling + WA-like	266	100	90	91	91	93	86
Kling + YT-like	266	100	89	87	88	79	76
OpenSora	267	100	96	98	98	98	97
OpenSora + WA-like	267	100	97	100	100	99	99
OpenSora + YT-like	267	100	98	100	100	100	100
Sora	49	100	**49**	**37**	**43**	**37**	**37**
Sora + WA-like	49	100	**51**	**47**	**57**	**41**	**37**
Sora + YT-like	49	100	**59**	**55**	**55**	**41**	**51**

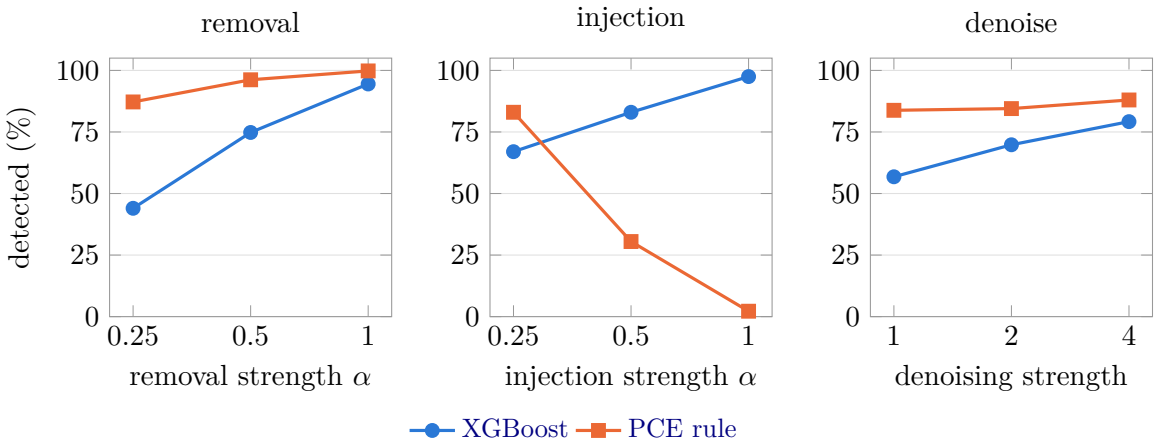

Figure 1: Attack strength vs detection (VISION attacks, realistic, 5-fold). The PCE rule "detects" removal only because low PCE is its sign of non-genuine; it misses injection, where the victim's fingerprint is matched.

6 Cross-dataset test

FloreView cameras are split by their PRNU baseline: 10 of 45 cameras have a usable video PRNU (attribution $\geq 50\%$); the others are modern phones whose stabilisation removes it.

Table 7: Cross-dataset detection rate (%), models trained on VISION + ACID + GenVidBench (fixed split).

Condition	n	PCE	SVM	XGB	GRU	CNN	Transf.
FloreView genuine, PRNU removed (stabilised)	1361	0	29	44	51	44	38
FloreView genuine, PRNU usable	355	62	70	78	79	69	80
GenBuster AI	616	100	68	74	74	71	62
GenBuster AI + WA-like	616	100	43	43	49	47	45
GenBuster AI + YT-like	616	100	65	76	67	70	64

7 Feature ablation (RQ1)

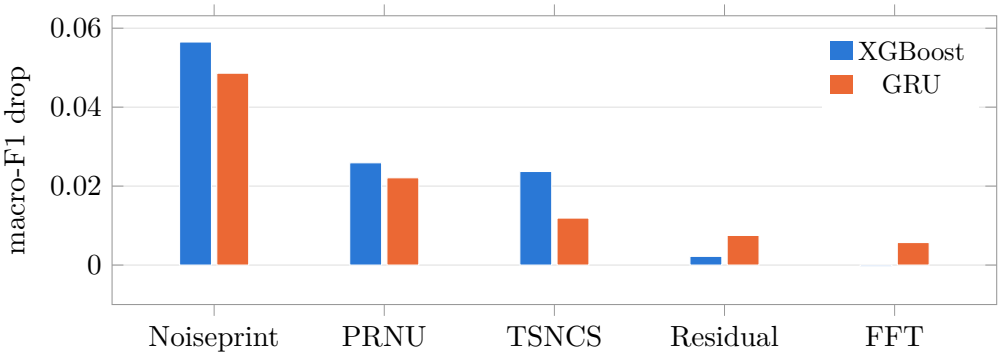

Figure 2: Drop in macro F1 when one feature group is removed (realistic, 5-fold, 5 frames). Larger = more important.

Table 8: Macro-F1 drop per feature group and model (5-fold, 5 frames).

Setup	Removed group	SVM	XGB	GRU	CNN	Transf.
realistic	Noiseprint	0.053	0.057	0.049	0.043	0.054
realistic	PRNU	0.021	0.026	0.022	0.009	0.016
realistic	TSNCS	0.012	0.024	0.012	0.016	0.018
realistic	Residual	0.002	0.002	0.008	-0.004	0.002
realistic	FFT	-0.001	-0.000	0.006	-0.003	0.003
controlled	Noiseprint	0.054	0.045	0.054	0.047	0.043
controlled	PRNU	0.031	0.033	0.025	0.020	0.017
controlled	TSNCS	0.014	0.024	0.013	0.011	0.002
controlled	Residual	0.009	0.004	0.001	-0.008	-0.006
controlled	FFT	-0.003	0.002	-0.001	0.005	0.005

8 Comparison with the base papers

The base papers solve related but different tasks, so each is compared on its own terms. Literature values are taken from the papers as reported; protocols differ (noted per table).

8.1 Camera identification: PDC-ViT (Elharrouss et al. 2025)

PDC-ViT is a learned camera-identification model (80/20 train-test split, every camera seen in training). The closest component of HVPF is the training-free PRNU baseline (Step 4): fingerprints from flat videos, tested on natural videos, closed-set over all cameras of the dataset; verification at $\mathrm{PCE}_0 > 60$.

Table 9: Camera identification. Accuracy = closed-set attribution; FNR/FPR = verification.

Method	Dataset (cameras)	Accuracy (%)	FNR (%)	FPR (%)	EER (%)
PDC-ViT [1]	VISION (35)	94.30	–	–	–
HVPF PRNU	VISION (35)	73.4	36.2	0.00	18.3
HVPF PRNU	VISION (21 non-stabilised)	100.0	–	–	–
PRNU [1]	Video-ACID + WA (26)	–	41.9	3.6	–
WA-PRNU [1]	Video-ACID + WA (26)	–	28.06	2.8	–
PDC-ViT [1]	Video-ACID + WA (26)	74.77	23.4	0.92	–
HVPF PRNU	Video-ACID (44)	59.1	59.6	0.00	25.5
HVPF PRNU	FloreView (45)	26.2	80.8	0.00	38.8

PRNU matching needs no training and is perfect on non-stabilised VISION cameras, above PDC-ViT; electronic stabilisation is what lowers the all-camera figures. Our verification threshold is strict (FPR 0%), so FNR is higher than PDC-ViT's (FPR 0.92%); the EER gives a threshold-free view. PDC-ViT was not tested on manipulated or synthetic video, and no base paper uses FloreView.

8.2 PRNU-transfer attack: Khan et al. 2025

Khan et al. inject a camera's PRNU into images from another camera or from AI generators and report that 85.5% of them pass source verification (images only). Our injection attack is the same attack on video. The compromise rate is the share of injected clips accepted as genuine: by the PCE rule if the victim's PCE_0 exceeds 60, by a model if it predicts *organic* (realistic, 5-fold, 5 frames).

Table 10: Compromise rate of PRNU injection (%, lower is better for the defender).

	Injection strength α			α=0.5, re-shared		
Detector	0.25	0.5	1	WA-like	YT-like	All
Khan et al. 2025 [2], images			–			85.5
PCE rule	17	70	98	2	26	42
SVM	26	22	8	20	25	20
XGBoost	31	17	2	26	30	21
GRU	30	18	4	21	27	20
1D-CNN	30	23	7	26	28	23
Transformer	32	20	4	18	29	20

A full-strength injection fools the PCE rule in 98% of the clips, in line with Khan et al.; the best model lets through only 2.5%. Weak injections (α=0.25) rarely fool PCE but are also harder for the models to spot.

8.3 Low false-positive operation: Bao et al. 2024

Bao et al. detect frame deletion (a different manipulation) from inter-frame noise consistency, the idea behind TSNCS, and stress performance at low false-positive rates; their results are given as ROC curves only. Table 11 reports HVPF at the same kind of operating points: share of non-genuine clips detected while at most 1, 5 or 10% of genuine clips are flagged.

Table 11: Detection rate of non-genuine clips at fixed false-positive rates (realistic, 5-fold, 5 frames).

Detector	FPR 1%	FPR 5%	FPR 10%
PCE rule	6.3	11.7	19.2
SVM	25.0	48.0	60.3
XGBoost	41.2	56.5	67.7
GRU	36.3	55.5	66.3
1D-CNN	29.3	50.5	62.4
Transformer	30.7	51.4	62.6

8.4 What the base papers do not cover

- PRNU injection and removal attacks on *video*, including platform-like re-sharing (Khan et al. test images only).
- Camera identification under attack (PDC-ViT reports clean data only).
- FloreView as an unseen-camera test set.
- One model that separates genuine, manipulated and AI-generated video.

9 Limitations and next steps

- Phones whose electronic stabilisation removes the PRNU (most FloreView cameras) are the hardest genuine case; more such phones in training is the most promising improvement.
- Platform-like copies make attacks much harder to see; detection of attacked clips shared through WhatsApp-like re-encoding is lower than for the original attacked clips (Table 6).
- Sequence models can learn the clip length; equal-length inputs (e.g. cyclic padding) would remove this.
- Sora is represented by only 51 clips; more high-quality modern generators are needed.

References

[1] O. Elharrouss et al., "PDC-ViT: source camera identification using pixel difference convolution and vision transformer," *Neural Computing and Applications*, 2025, doi:10.1007/s00521-025-11004-z.

[2] Khan, Hasan, Islam, "Spoofing camera source attribution via PRNU transfer attacks," *ISC 2025*, Springer LNCS 16186, pp. 279–299, 2025.

[3] Bao et al., "Anti-forensics video forgery detection via noise transfer matrix," *Sensors* 24(16):5341, 2024, doi:10.3390/s24165341.